

Beyond the Ironies: Tracing the Evolution of Human-Computer Collaboration in Industrial AI Since 1983

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Abstract—This paper discusses the ways in which the automation of industrial processes through Artificial Intelligence (AI) and Industry 5.0 paradigms may expand rather than eliminate problems with the human operator. Some comments will be made on methods of alleviating the “classic” ironies of automation, first identified in 1983, within the context of modern predictive maintenance and Augmented Reality (AR) systems. It is suggested that while the technological landscape has shifted from rigid control to probabilistic Machine Learning (ML), the fundamental paradox remains: the more advanced a control system is, so the more crucial may be the contribution of the human operator. Furthermore, the introduction of Generative AI (GenAI) creates a new irony, in which the operator is relieved of content generation only to be burdened with the more complex cognitive task of output evaluation. The conclusion is drawn that automated systems still are man-machine systems, for which human-computer collaboration is a requirement for system robustness.

Index Terms—Ironies of Automation, Human-Robot Collaboration (HRC), Industry 5.0, Digital Twins, Predictive Maintenance, Generative AI, Augmented Reality, Human-Centric Design.

I. INTRODUCTION

The classic aim of automation is to replace human manual control, planning and problem solving by automatic devices and computers. The prevailing logic has long been that the human operator is unreliable, inefficient and prone to fatigue, so should be eliminated from the system to ensure stability and precision. However, as pointed out in previous works (Bainbridge, 1983), there is a major irony in this approach: the designer who tries to eliminate the operator still leaves the operator to do the tasks which the designer cannot think how to automate. This paper suggests that the increased interest in human factors among engineers in the era of Industry 4.0 and 5.0 reflects the persistence of this irony. The result of such automation is not the removal of the human element, but a shift in the nature of the operator’s burden from direct manipulation to the more cognitively demanding tasks of supervision, adjustment, and intervention during abnormal conditions.

This paper is particularly concerned with the evolution of control in process industries and the subsequent shift towards ‘human-centric’ design. To understand this evolution, one must distinguish between the paradigms of Industry 4.0 and Industry 5.0. As Rashid and Kausik (2024) note, the

foundations of Industry 4.0 are Artificial Intelligence (AI), Big Data, the Internet of Things (IoT) and Cyber-Physical Systems (CPS). The primary objective of this fourth industrial revolution was the optimisation of manufacturing processes through interconnectivity and data autonomy, often viewing the human worker as a source of variance to be minimised. In contrast, Industry 5.0 represents a philosophical correction to this technocratic focus. As Martini and colleagues (2024) articulate, Industry 5.0 is not merely about production efficiency but about placing the welfare of the worker at the centre of the process, emphasising resilience and sustainability alongside throughput. It is suggested that this transition acknowledges a fundamental truth ignored by earlier automation efforts: that the flexibility and adaptability of the human operator are resources that cannot be fully replicated by rigid control logic.

Despite the technological leaps made since 1983, the vision of ‘lights-out manufacturing’—a production environment where factories operate autonomously without human presence—has not been realised. In the ‘classic’ view, the designer’s view of the human operator may be that the operator is a liability. Yet, even highly automated systems, such as those driven by modern Artificial Intelligence (AI) and Digital Twins (DT), need human beings for supervision and to handle the edge cases that algorithms cannot predict. If the automated system encounters a situation for which it has not been programmed, it will either halt or fail; it is the human operator who must step in to bridge the gap between the system’s logic and the reality of the plant. Therefore one can draw the paradoxical conclusion that automated systems still are man-machine systems, for which human-computer collaboration is a requirement for system robustness. The structure of this paper traces this evolution. The first sections discuss the persistence of the original ironies regarding manual and cognitive skills in the context of modern algorithms, utilising foundational work on the subject (Bainbridge, 1983). The later sections introduce some aspects of human-computer collaboration, specifically how technologies such as predictive maintenance and Augmented Reality (AR) may alleviate the problems of the operator who is expected to take over in abnormal conditions. Finally, the paper examines the emergence of Generative AI (GenAI), as discussed by Simkute and colleagues (2024), which presents a new irony: the operator is relieved of content generation only to be burdened with

the more complex cognitive task of output evaluation. It is concluded that as systems become more advanced, the consequences of the ironies of automation do not disappear; they merely oscillate into new forms.

II. THE PERSISTENCE OF BAINBRIDGE'S IRONIES: A LITERATURE REVIEW

To understand the challenges facing Industry 5.0, one must first revisit the foundational critiques of automation that emerged during the previous industrial transitions. The seminal work of Bainbridge (1983) remains the touchstone for this analysis. The important ironies of the classic approach to automation lie in the expectations of the system designers, and in the nature of the tasks left for the human operators to carry out. In 1983, the prevailing attitude—often summarised by the 'Fitts List'—was to assign to the machine what it does best (speed, power, computation) and to the man what he does best (heuristics, flexibility), often leaving the operator with an arbitrary collection of tasks. This paper suggests that despite the passage of four decades and the introduction of advanced Artificial Intelligence (AI), the fundamental paradox remains: the more advanced a control system is, so the more crucial may be the contribution of the human operator.

A. The 'Impossible Task' of Monitoring

It may seem that the operator who is expected solely to monitor that the automatics are acting correctly has a relatively simple task. The logic of the designer is often that by removing the manual burden of control, the operator is freed to supervise the process with greater clarity. However, this view fundamentally misunderstands human physiology. As pointed out in previous works (Bainbridge, 1983), referencing the earlier vigilance studies of Mackworth (1950), it is impossible for even a highly motivated human being to maintain effective visual attention towards a source of information on which very little happens, for more than about half an hour. The human monitor has been given an impossible task.

Strauch (2018) reinforces this, arguing that the ironies are still unresolved in modern aviation and industrial contexts. When a system is highly reliable, the operator is lulled into a state of reduced vigilance. Consequently, when an anomaly finally occurs, the operator is physically and cognitively unprepared to detect it. There is some concern that if the operator is merely a monitor, he will not be able to generate successful new strategies for unusual situations. The designer puts the human in the loop to handle the rare, unpredictable failures, yet the very act of automating the routine operations deprives the operator of the situational engagement required to handle those failures.

B. The Decay of Manual and Cognitive Skills

There are two general categories of task left for an operator in an automated system: manual control and cognitive skills. Unfortunately, physical skills deteriorate when they are not used. This means that a formerly experienced operator who has been monitoring an automated process may now be an

inexperienced one. If he takes over, he may set the process into oscillation. As Edwards and Lees (1974) have shown, the experienced operator makes the minimum number of actions, while the inexperienced one oscillates round the target value.

This degradation is not limited to physical manipulation; it extends to the cognitive understanding of the process. An operator who finds out how to control the plant for himself uses a set of propositions about possible process behaviour. However, this type of knowledge develops only through use and feedback. Simkute and colleagues (2024) identify a similar phenomenon in the context of Generative AI, terming it 'task-complexity polarisation'. Automation tends to make easy tasks easier while making hard tasks harder. By automating the routine interactions, the system denies the operator the opportunity to practise solving small problems within known information. Therefore, when the system fails, it invariably does so in complex, non-routine ways that require a depth of understanding the operator has been prevented from maintaining.

Baxter and colleagues (2012) note that modern interfaces, such as 'glass cockpits' and abstract cloud computing dashboards, further exacerbate this issue by hiding the complexity of the underlying system. The irony is that the technology helps the operator develop a mental model of the situation, but these models may be inconsistent with the state of the world because the automation does not present the operator with the information he needs in an obvious and timely manner. The operator is thus left with a 'surface' understanding of the system, which is sufficient for normal operation but catastrophically inadequate for fault diagnosis.

C. Trust, Complacency, and the Operator's Dilemma

The relationship between the operator and the machine is mediated by trust. Frateczak and colleagues (2021) cite Mayer and colleagues (1995) in defining trust as 'the willingness of a party to be vulnerable to the actions of another party'. In an industrial context, this manifests in two distinct failure modes: over-trust (complacency) and under-trust (scepticism).

If the automated system is highly reliable, the operator may develop over-trust, leading to complacency. As noted in the discussion of vigilance, the operator assumes the system will handle all variances and ceases to critically evaluate its output. This is particularly dangerous in systems described by Strauch (2018), such as the Air France 447 crash, where the pilots, conditioned by the reliability of the autopilot, failed to recognise that the system had disengaged and handed control back to them during a stall event. The operator needs to be more rather than less skilled, yet the automation has deprived him of the opportunity to maintain that skill.

Conversely, under-trust leads to inefficiency. If a system is perceived as unreliable, or if its decision-making logic is opaque ('black box'), the operator may choose to ignore its recommendations or disable it entirely. Frateczak and colleagues (2021) suggest that trust is dynamic and can be repaired; for instance, a robot that 'apologises' or communicates its error status clearly can help maintain the operator's trust.

However, without such mechanisms, the designer's intent to optimise the process is thwarted by the operator's refusal to rely on the tool.

D. Situation Awareness and the 'Out-of-the-Loop' Problem

The culmination of these factors is the loss of Situation Awareness (SA). Simkute and colleagues (2024), drawing on Endsley (1995), describe how removing the human from active production reduces his awareness of the system state. In a manual system, the operator receives constant feedback: vibration, sound and resistance. In an automated system, this feedback is filtered or removed.

If the decisions can be fully specified, then a computer can make them more quickly, taking into account more dimensions. There is therefore no way in which the human operator can check in real-time that the computer is following its rules correctly. The operator is forced to monitor the system at a meta-level, checking the output rather than the process. When the output deviates from the expected norm, the operator must orient himself, determine the state of the system, and formulate a plan—all under the pressure of a failure scenario.

One might state the current situation as a paradox: by automating the process to a higher degree, the industry has realised the necessity of the human operator, yet has created an environment where the human is least able to succeed. The designer who tries to eliminate the operator still leaves the operator to do the tasks which the designer cannot think how to automate, specifically the handling of unforeseen variances. The persistence of these ironies suggests that simply adding more automation or 'smarter' AI will not solve the problem; rather, it requires a fundamental rethinking of the human-computer collaboration model, moving from rigid allocation of tasks to a more symbiotic relationship.

III. FROM RIGID LOGIC TO PROBABILISTIC UNCERTAINTY

Since 1983, the nature of automatic devices has changed fundamentally. We have moved from the 'classic' aim of rigid, rule-based automation to the era of Industry 4.0 and 5.0, where the underlying logic of control has shifted from determinism to probability. As Rashid and Kausik (2024) point out, the evolution of Artificial Intelligence (AI) has shifted from symbolic systems—logic-based structures defined by explicit 'If X, then Y' rules—to connectionist systems involving Neural Networks and Machine Learning (ML). In process plants, the different modes of operation are now supported by data-driven modelling, where algorithms learn from massive multi-source datasets. This contrasts sharply with the classic approach where the designer tried to specify all process behaviours in advance, creating a system that was brittle but predictable.

A. The Transparency Paradox

In the era of Bainbridge, the designer who tried to eliminate the operator created systems of rigid control. These systems were inflexible and often failed in the face of unanticipated

variance, yet they possessed a quality of transparency; if the inputs were known, the output was deterministic and traceable. In contrast, modern Deep Learning (DL) methods, as reviewed by Drakaki and colleagues (2022), handle complexity by extracting features from raw data without manual intervention. This allows for the automation of tasks previously thought to require human intuition, such as acoustic fault diagnosis or visual inspection. However, this flexibility comes at the high cost of explainability. The internal logic of a Neural Network is often a 'black box' to the human operator. Therefore one can draw the paradoxical conclusion that while the system is more capable of handling variance, it is less capable of explaining why it acted in a specific manner to the human supervisor. The operator is presented with a decision—such as a recommendation to shut down a turbine—derived from millions of parameters within a hidden layer, leaving him with no way to verify the computer's reasoning in real-time.

B. The Conflict Between Safety and Probability

This technological shift introduces a significant conflict within the safety-critical context of industrial operations. Traditional industrial safety relies on certainty; a safety interlock must trigger if a specific pressure threshold is crossed, without ambiguity. However, probabilistic Machine Learning models operate on confidence intervals rather than absolute truths. Chen and colleagues (2023) note that while data-driven models can predict Remaining Useful Life (RUL) with high accuracy, they rely on historical data which may not encompass every possible failure mode. A model may predict a failure with 95% confidence, but the industrial environment requires a binary decision: run or stop.

If the computer is being used to make decisions because human judgement is not adequate in this context, then the computer should also present the information in a way the operator can follow. Yet, if the system presents a recommendation with only statistical certainty, the operator is left with the burden of evaluating the remaining margin of uncertainty, often under extreme time pressure. This effectively transfers the risk of the algorithmic error to the human operator, who must decide whether to trust a probabilistic output that he cannot fully interrogate.

C. Supervising the Learning Machine

Finally, the introduction of adaptive or 'on-line' learning systems alters the fundamental stability of the man-machine system. In a static automation environment, the operator's mental model of the plant, once formed through training and experience, remains valid over time. In a system driven by on-line learning, the behaviour of the machine changes as it ingests new data and updates its weights. Simkute and colleagues (2024) highlight a similar issue in the context of Generative AI, noting that non-determinism produces different outputs for the same input, resulting in lower reliability from the user's perspective.

For the industrial operator, this implies that a control strategy that stabilised the plant yesterday may set the process

into oscillation today if the underlying algorithm has drifted or adapted to a new local minimum. The operator is thus required to supervise a moving target, constantly updating his mental model to match an algorithm that is itself in flux. There is some concern that this dynamic creates a new form of the vigilance problem identified by Mackworth (1950): the operator must not only monitor for rare failures but must also detect subtle shifts in the automation's behaviour that may indicate model drift. The designer has thus replaced the physical fatigue of manual control with the cognitive exhaustion of supervising a system that is essentially unknowable.

IV. EMPIRICAL EVIDENCE AND TECHNOLOGICAL SOLUTIONS

The theoretical concerns regarding the ironies of automation are not merely academic abstractions; they have manifested repeatedly in catastrophic system failures across various industries. One might state these problems as a paradox: by automating the process to improve safety and efficiency, the industry has created a class of accidents which are only possible because of the automation. This section examines empirical evidence from aviation and automotive sectors to illustrate the persistence of “clumsy automation,” and subsequently discusses how emerging technologies in Industry 5.0 may offer methods of alleviating—though not necessarily curing—these fundamental ironies.

A. The Persistence of “Clumsy Automation”: Case Studies

The aviation industry, having been at the forefront of automation for decades, provides the clearest evidence of the warnings regarding the decay of manual skills and the loss of situation awareness. Strauch (2018) argues that the ironies remain unresolved, citing the crash of Air France Flight 447 in 2009 as a definitive example. In this incident, the aircraft's automated systems disengaged due to ice obstructing the pitot tubes, a sensor failure that deprived the flight computer of airspeed data. The pilots, suddenly handed manual control of the aircraft at high altitude and in darkness, failed to recognize the stall warnings and maintained a nose-up input, causing the aircraft to descend into the ocean.

This accident illustrates two critical ironies. First, the decay of manual skills: the pilots were highly experienced in monitoring the flight management system but lacked recent practice in manual handling at high altitudes, a task the automation had effectively usurped. Second, mode confusion: the complexity of the interface obscured the fact that the protection systems (which normally prevent a stall) were no longer active. Baxter et al. (2012) reinforce this, noting that “glass cockpits” hide the complexity of the underlying system, making it difficult for the operator to interpret whether the feedback shows that there is something wrong with the system or more simply that he has misjudged the situation. The pilots were victims of a system designed to handle 99% of operations, leaving them unprepared for the 1% of variance that required human intuition.

A similar irony is currently unfolding in the automotive sector with the deployment of semi-autonomous driving systems, such as Tesla's Autopilot. This represents the “Impossible Task” of monitoring in its most acute form. The driver is expected to supervise a vehicle that performs correctly for hundreds of miles, lulling them into a state of complacency. Yet, they must be ready to intervene within a fraction of a second should the computer misinterpret a visual cue. As noted in the discussion of Mackworth (1950), it is impossible for a human being to maintain effective vigilance towards a source of information on which very little happens. Strauch (2018) highlights that while automation in vehicles aims to reduce driver error, it introduces new failure modes associated with the “distracted operator.” The irony is compounded by the marketing of these systems as “autonomous,” which encourages over-trust, while the engineering reality requires a level of supervision that the human brain is ill-equipped to provide during passive monitoring.

B. Alleviating the Ironies: The Role of Industrial AI

If the fundamental problem is that the operator is burdened with tasks they are ill-suited to perform, then the solution must involve restructuring the human-machine interaction rather than simply increasing the level of automation. Recent developments in predictive maintenance (PdM), Digital Twins (DT), and Augmented Reality (AR) offer potential pathways for this restructuring.

1) *Transforming Vigilance into Analysis: Predictive Maintenance:* A serious irony identified by Bainbridge was that the operator is asked to monitor the automatics for rare abnormalities—a task requiring sustained vigilance. AI-driven Predictive Maintenance (PdM) offers a method of alleviating this by shifting the operator's role from real-time monitoring to strategic analysis.

As Chen et al. (2023) explain, modern PdM utilizes Deep Learning (DL) algorithms to extract features from raw data—such as vibration signals or acoustic patterns—more effectively than a human monitor could. Instead of requiring the operator to watch a dial for a sudden spike (vigilance), the system calculates a Health Indicator (HI) and predicts the Remaining Useful Life (RUL) of the asset. Mian et al. (2023) demonstrate that an Artificial Intelligence of Things (AIoT) framework can perform real-time anomaly detection at the edge, sending only high-level decision data to the cloud or the operator. This relieves the operator of the impossible task of vigilance. The human is no longer a passive monitor of a static process but an active analyst of a degradation trend, allowing for intervention to be planned rather than reactive. However, one must remain cautious; if the underlying ML model is opaque, the operator may struggle to trust the RUL prediction, returning to the dilemma of under-trust discussed in previous sections.

2) *Addressing Skill Degradation: Digital Twins and Simulation:* If the human operator is not involved in on-line control, he will not have detailed knowledge of the current state of the system, nor will he maintain the manual skills required for

intervention. The Digital Twin (DT) offers a partial solution to this decay of skills by providing a high-fidelity simulation environment.

Sajadieh and Noh (2025) describe the use of DTs to create virtualization and synchronization between the physical and digital worlds. This allows the operator to practice solving problems within the known information, and crucially, to generate strategies for low probability events which may be known to the trainer but not to the trainee. By running scenarios in the “virtual space”—such as a sudden pressure loss or a sensor failure—the operator can maintain cognitive and manual proficiency without risking the physical plant. This directly addresses Bainbridge’s concern that operators of automated systems cannot learn through feedback because the system corrects errors before they become visible. In the DT, the operator can be allowed to fail, thereby relearning the dynamics of the process that the automation usually hides.

3) *Mitigating Cognitive Load: Augmented Reality:* Finally, the opacity of modern systems creates a cognitive burden when the operator attempts to correlate system data with physical reality. Augmented Reality (AR) systems can reduce this load by presenting information in a way the operator can follow immediately.

Michalos et al. (2015) demonstrate how AR tools can support the operator in a shared workspace by displaying instructions and safety zones directly in the field of view. In a maintenance scenario, rather than consulting a manual or interpreting a 2D schematic on a separate screen, the operator sees the internal state of the machine superimposed on the physical asset. This addresses the problem that efficient retrieval of knowledge from long-term memory depends on frequency of use. By providing the information externally and contextually, the system reduces the load on the operator’s working storage. Furthermore, Mourtou et al. (2017) propose multimodal communication, where the robot communicates intent through natural language or visual cues, helping the operator maintain a “feel” for what the process is doing.

C. Conclusion on Technological Solutions

It is suggested that while these technologies do not eliminate the ironies of automation, they alter the nature of the man-machine relationship. PdM removes the need for impossible vigilance; DTs provide a sandbox for skill maintenance; and AR bridges the gap between the system’s logic and the physical world. However, the paradox remains: the more successful these systems are, the more rare the need for manual intervention becomes, and the more critical that intervention is when it finally occurs. The implementation of these technologies must therefore be driven not by a desire to eliminate the human, but to support the specific cognitive and physical needs of the operator who remains the final line of defense.

V. THE PARADOX OF GENERATIVE AUTOMATION

The classic aim of automation has historically been to replace human manual control and the closure of simple feedback loops by automatic devices. However, the recent

introduction of Generative AI (GenAI) suggests a new irony in the allocation of function between man and machine. While previous approaches sought to execute pre-defined rules, these new systems attempt to generate novel content and solutions. This paper discusses the ways in which this technological development may expand rather than eliminate the cognitive load on the human operator. It creates a particularly acute paradox: the operator is relieved of the task of content generation only to be left with the more complex cognitive task of output evaluation.

A. From production to evaluation

Simkute and colleagues (2024) have identified this phenomenon as a “production-to-evaluation shift.” In the classic manual approach, the operator or engineer constructs a solution step-by-step. Through this process of generation, he builds up a set of propositions about the system’s logic; he understands the function of a specific variable because he has placed it there himself. In a workflow mediated by generative automatics, the system produces the entire solution instantly. The operator’s role is transformed from that of an author to that of a critic.

The designer’s view of this shift may be that it represents a gain in productivity. However, this view ignores the nature of the cognitive tasks left for the operator. When an operator generates a schedule or code from scratch, he is engaged in a constructive process which naturally limits complexity to that which can be held in working storage. When an automatic system produces a solution, it provides a completed, complex artifact which the operator must reverse-engineer to understand. Consequently, the operator is faced with a task which is cognitively more difficult than the original work: he must identify subtle errors in a logic structure which he did not create.

B. Task complexity and logic

This dynamic is particularly visible in the domain of software engineering, which shares many characteristics with industrial logic design. Tools which allow for the rapid generation of code blocks lead, as noted by Simkute and colleagues (2024), to “task-complexity polarization.” The automation makes easy tasks—such as generating standard routines—easier, but it renders the diagnosis of complex logic significantly harder.

Just as manual operators lose the “feel” of the process dynamics when removed from the control loop, operators using generative tools lose the rich feedback associated with writing logic line-by-line. They are presented with a “wall of code” which may appear syntactically correct but contains subtle failures in logic. The irony is that to effectively evaluate the computer’s output, the operator needs a level of expertise higher than that required to perform the task manually. If the operator relies on the automatics because he lacks the skill to generate the solution, he is by definition unqualified to evaluate whether the computer’s decision is acceptable.

The designer who implements such systems to assist less-experienced workers may therefore be creating a man-machine system where those workers are most likely to fail.

C. Hallucinations and monitoring

A further danger lies in the phenomenon of “hallucination,” where the generative model produces plausible but factually incorrect information. In an industrial context, the consequences of such errors are not merely inconvenient; they are potentially hazardous. Consider an operator interacting with a maintenance assistant. The computer, drawing on probabilistic associations rather than a deterministic knowledge of plant physics, might suggest a procedure which sounds authoritative but references a valve which does not exist or bypasses a safety interlock.

This reintroduces the trust dilemma discussed in classic automation studies. If the system is generally reliable, the operator will not monitor the automatics effectively, and may accept the hallucinated procedure without verification. Conversely, if the operator catches the system in a fabrication, his trust may collapse, leading him to abandon a tool which could be useful. The operator is thus placed in a double bind: he must use the tool to maintain efficiency, yet he cannot fully trust its output. The burden of filtering truth from hallucination falls entirely on the human, transforming the operator into a safety monitor for the algorithm’s uncertainties.

D. The erosion of skill

Finally, one must consider the long-term impact of this paradigm on the development of operator knowledge. There is a concern that if the “easy” tasks are permanently delegated to the automatics, the pathway for junior operators to develop expertise will be severed. Long-term knowledge develops only through use and feedback about its effectiveness; expertise is often built through the repetitive performance of routine tasks.

If a junior engineer never generates the basic logic for a controller because the computer generates it for him, he may never develop the deep, intuitive understanding of the system required to diagnose a critical failure. He becomes, effectively, an inexperienced supervisor of a highly complex system. This echoes the warning regarding the decay of manual skills, transposed here into the cognitive domain. The final irony may be that the extensive use of generative systems to support the workforce eventually erodes the very human intelligence required to train and supervise the computer. The industry risks creating a generation of operators who can operate the interface but do not understand the machine, leaving the system fragile in the face of abnormal conditions.

VI. CONCLUSION

This paper has discussed the evolution of the “classic” ironies of automation through the technological paradigms of Industry 4.0 and the emerging landscape of Industry 5.0. Although the specific mechanisms of control have shifted from rigid electromechanical logic to probabilistic artificial intelligence, the fundamental paradox remains unresolved: the

designer who tries to eliminate the operator still leaves the operator to do the tasks which the designer cannot think how to automate.

We have moved from the ironies of manual control to the ironies of supervision, and now, with the advent of generative systems, to the ironies of evaluation. In the classic approach, the primary concern was the deterioration of manual skills and the impossibility of maintaining vigilance over static displays. With the introduction of machine learning, the burden shifted to interpreting opaque, probabilistic outputs from “black box” systems. Now that generative systems produce code and logic instantly, the operator is relieved of the effort of creation only to be burdened with the cognitively more difficult task of verifying complex, machine-generated outputs for subtle errors. The task has shifted from “doing” to “watching”, and finally to “critiquing”, yet in each stage the system relies on a human operator who is increasingly detached from the underlying process to intervene when the automatics fail.

It is suggested that the technological solutions reviewed in this paper—predictive maintenance, digital twins, and augmented reality—offer methods of alleviating these problems, but they do not eliminate them. Predictive maintenance transforms the impossible task of vigilance into a manageable task of analysis, yet it requires trust in probabilistic models. Digital twins provide a simulation environment to maintain skills which would otherwise deteriorate, yet they require significant investment to ensure the virtual model remains synchronised with physical reality. Augmented reality reduces working storage load by presenting information in the visual field, yet it risks creating a dependency where the operator cannot function without the overlay.

Therefore one can draw the conclusion that the shift towards “human-centric” design is not merely an ethical preference; it is a functional requirement for system safety. The aim of “lights-out” manufacturing ignores the reality that automated systems are brittle in the face of unanticipated variance. It is the human operator’s flexibility and ability to reason through uncertainty which bridges the gap between the rigid logic of the algorithm and the chaotic reality of the physical world.

The ingenious suggestions reviewed in this paper show that humans working without time-pressure can be impressive problem solvers. The difficulty remains that they are less effective when under time pressure during a system failure. Consequently, the industry must move beyond the “Fitts List” mentality of assigning tasks based on raw capability. Instead, the design of automated systems must prioritize the maintenance of the operator’s situation awareness and skill level. Perhaps the final irony is that it is the most successful, most advanced automated systems which need the greatest investment in human operator training and support, for it is only the skilled human who can save the machine when it inevitably falters.

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